

DINWIDDIE, VA, USA

WMO: 724014

Lat: 37.183N

Lon: 77.500W

Elev: 59

StdP: 100.62

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 93714

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-9.0	-7.1	-17.9	0.8	-3.5	-15.0	1.0	-1.5	9.0	8.4	8.2	8.1	1.0	350	0.410

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.4	36.2	24.9	34.1	24.4	32.9	23.8	27.0	32.9	26.1	32.1	25.4	30.9	2.9	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.4	20.8	29.6	24.2	19.2	28.5	23.8	18.8	28.2	84.7	32.9	81.2	32.2	77.8	31.1	29.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.9	6.9	5.7	-14.3	38.0	3.4	2.0	-16.7	39.5	-18.7	40.7	-20.6	41.9	-23.1	43.3		
(5)			-14.6	28.0	3.4	1.4	-17.0	29.0	-19.0	29.8	-20.9	30.6	-23.3	31.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	15.1	3.9	5.3	9.3	15.0	19.6	24.0	26.2	25.3	21.7	15.4	9.5	5.6
	DBStd	9.12	6.06	5.37	5.48	4.99	4.20	3.37	2.69	2.72	3.50	4.70	4.98	5.19
	HDD10.0	681	207	149	81	11	0	0	0	0	0	8	68	157
	HDD18.3	2051	449	365	284	125	38	3	0	0	11	116	266	395
	CDD10.0	2556	17	18	60	161	298	421	504	474	352	175	54	22
	CDD18.3	883	1	1	4	26	77	173	246	216	112	24	2	1
	CDH23.3	9356	3	4	63	343	844	1847	2711	2220	1034	262	21	4
	CDH26.7	4158	1	0	11	109	314	841	1364	1053	388	77	1	0
Wind	WSAvg	2.1	2.4	2.5	2.7	2.7	2.1	1.8	1.5	1.5	1.9	1.9	2.0	2.1
Precipitation	PrecAvg	1151	86	68	109	86	101	95	115	118	115	83	81	94
	PrecMax	1749	186	161	204	178	271	201	200	354	353	180	239	228
	PrecMin	869	22	20	6	33	28	23	32	33	11	0	5	18
	PrecStd	207	38	34	46	39	50	45	45	64	86	48	57	45
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	22.8	23.6	27.9	32.4	34.0	37.4	38.6	37.5	34.1	32.2	26.1	23.0
		MCWB	17.0	16.3	17.4	19.5	22.7	24.6	24.9	25.9	23.7	22.4	18.1	18.2
	2%	DB	19.5	21.1	25.1	29.6	32.2	35.0	36.4	35.4	32.5	28.6	23.0	20.3
		MCWB	15.6	14.0	15.6	18.7	22.4	24.6	25.3	24.9	23.2	21.1	16.6	16.9
	5%	DB	17.3	17.8	22.5	27.5	30.2	32.9	34.3	33.7	31.0	26.3	21.3	17.5
		MCWB	13.7	12.4	14.7	17.8	21.7	24.0	24.8	24.3	22.8	19.7	16.5	13.9
	10%	DB	13.9	15.3	19.6	24.9	27.9	31.6	32.8	32.2	28.8	23.8	18.8	14.9
		MCWB	10.3	10.1	13.5	16.4	20.4	23.4	24.3	23.8	22.0	18.4	15.1	11.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	18.8	18.1	19.4	22.5	25.1	26.9	28.3	27.9	26.3	24.3	21.2	19.8
		MCDB	21.1	21.6	24.4	29.7	31.4	33.3	34.2	34.0	31.1	28.1	23.2	21.5
	2%	WB	16.7	15.9	17.7	20.6	23.8	26.0	27.2	26.8	25.1	22.6	19.1	17.4
		MCDB	18.9	19.3	21.9	26.5	29.4	32.4	32.9	32.7	29.1	26.4	21.5	19.3
	5%	WB	14.2	13.6	16.4	19.3	22.6	25.3	26.3	25.8	24.1	21.3	17.4	14.9
		MCDB	17.0	17.1	21.6	24.6	28.1	31.2	32.4	31.4	28.1	25.1	19.5	17.3
	10%	WB	10.9	10.8	14.4	17.9	21.7	24.4	25.5	25.1	23.2	19.6	15.5	11.9
		MCDB	13.7	14.9	18.7	22.7	26.8	29.5	31.1	30.1	26.9	22.9	18.6	13.9
Mean Daily Temperature Range	5% DB	MDBR	11.8	12.8	13.6	14.4	13.4	12.6	12.4	12.3	12.5	13.7	13.7	12.0
		MCDBR	15.2	17.1	17.0	16.7	15.6	14.6	14.3	14.1	14.6	15.9	16.2	14.6
		MCWBR	11.2	11.5	9.4	7.7	7.5	6.4	5.6	5.6	6.5	8.8	10.3	10.9
	5% WB	MCDBR	13.8	15.1	14.0	14.3	13.3	12.7	12.3	12.2	11.4	12.8	12.4	12.4
		MCWBR	11.3	11.7	9.3	7.9	6.9	5.9	5.7	5.7	5.9	7.6	9.4	10.4
Clear-Sky Solar Irradiance	taub	0.299	0.313	0.345	0.388	0.428	0.485	0.506	0.507	0.404	0.354	0.319	0.303	
	taud	2.535	2.489	2.435	2.328	2.241	2.128	2.090	2.071	2.350	2.485	2.537	2.559	
	Ebn at Noon	898	925	920	893	858	807	786	775	846	865	865	869	
	Edh at Noon	77	91	106	125	139	155	160	160	115	91	77	71	
All-Sky Solar Radiation	RadAvg	2.36	3.11	4.09	5.17	5.84	6.33	6.02	5.45	4.44	3.50	2.64	2.08	
	RadStd	0.19	0.35	0.33	0.41	0.51	0.42	0.30	0.28	0.44	0.43	0.21	0.18	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (73 neighbors)	+0.42	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+149	+97

Nomenclature: See separate page